

Sec 3.3 – The Product and Quotient Rules

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

In Chap 3, each section introduces a new rule, or shortcut, to differentiation. So as we learn new rules, we still want to be aware of the rules we've learned previously. And we will be able to calculate many of our derivatives multiple ways.

For example, we can do the following derivative with what we learned in the previous sections.

$$\text{Let } g(x) = \frac{x^2 + 3}{x}. \text{ Find } g'(x).$$

We can divide each term of the numerator by x to rewrite:

$$g(x) = \frac{x^2 + 3}{x} = \frac{x^2}{x} + \frac{3}{x} = x + 3x^{-1}. \text{ Then } g'(x) = 1 - 3x^{-2}$$

But suppose we want to find derivatives of the following:

$$f(x) = \frac{x^2 + 3x}{x^3 + 4} \text{ or } h(t) = \frac{2^x}{x^3 - 2x}$$

In these cases we are not able to divide, so we need a new way to deal with quotients of functions!

Theorem

If f and g are differentiable functions, then $\frac{f}{g}$ is differentiable whenever $g(x) \neq 0$ and

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$$

Use the Quotient Rule to find the following derivatives:

$$\begin{aligned} \textcircled{1} \left(\frac{x^2 + 3}{x}\right)' &= \frac{(x^2 + 3)'(x) - (x^2 + 3)(x)'}{(x)^2} = \frac{2x(x) - (x^2 + 3)(1)}{x^2} \\ &= \frac{2x^2 - x^2 - 3}{x^2} = \frac{x^2 - 3}{x^2} = 1 - \frac{3}{x^2} \end{aligned}$$

$$\begin{aligned} \textcircled{2} \left(\frac{x^2 + 3x}{x^3 + 4}\right)' &= \frac{(x^2 + 3x)'(x^3 + 4) - (x^2 + 3x)(x^3 + 4)'}{(x^3 + 4)^2} \\ &= \frac{(2x + 3)(x^3 + 4) - (x^2 + 3x)(3x^2)}{(x^3 + 4)^2} \end{aligned}$$

$$\begin{aligned} \textcircled{3} \left(\frac{2^x}{x^3 - 2x}\right)' &= \frac{(2^x)'(x^3 - 2x) - (2^x)(x^3 - 2x)'}{(x^3 - 2x)^2} \\ &= \frac{(\ln 2)2^x(x^3 - 2x) - 2^x(3x^2 - 2)}{(x^3 - 2x)^2} \end{aligned}$$

1. Suppose $f(3) = 2$ and $f'(3) = -4$ and let $F(x) = \frac{f(x)}{x^3}$. Find $F'(3)$.
(Hint: Write out the quotient rule for $F'(x)$.)
2. Find the equation of the tangent line to $f(x) = \frac{x^2 + 2}{x + 4}$ at $x = -2$.
3. Find $\frac{dy}{dx}$ when $y = \frac{\sqrt[3]{x} + 4x^2}{3x^{3/2} + x^3}$

1. Suppose $f(3) = 2$ and $f'(3) = -4$ and let $F(x) = \frac{f(x)}{x^3}$. Find $F'(3)$.
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Solution: By the quotient rule,

$$F'(x) = \frac{(f(x))'(x^3) - f(x)(x^3)'}{(x^3)^2} = \frac{f'(x)x^3 - f(x)(3x^2)}{x^6}. \text{ Then}$$

$$F'(3) = \frac{f'(3)3^3 - f(3)(3(3)^2)}{3^6} = \frac{(-4)(27) - 2(27)}{3^6} = -\frac{2}{9}$$

2. Find the equation of the tangent line to $f(x) = \frac{x^2 + 2}{x + 4}$ at $x = -2$.

Solution: The equation of the tangent line will have the form

$$y = m(x - a) + b, \text{ where } a = -2 \text{ and } b = f(-2) = \frac{4 + 2}{-2 + 4} = 3. \text{ To}$$

find the slope, m , we take the derivative:

$$\begin{aligned} f'(x) &= \frac{(x^2 + 2)'(x + 4) - (x^2 + 2)(x + 4)'}{(x + 4)^2} \\ &= \frac{(2x)(x + 4) - (x^2 + 2)(1)}{(x + 4)^2}. \end{aligned}$$

Then

$$m = f'(-2) = \frac{(-2 + 4)(-4) - (4 + 2)}{(-2 + 4)^2} = \frac{-8 - 6}{4} = \frac{-14}{4} = -\frac{7}{2} \text{ and}$$

the equation of the tangent line is $y = -\frac{7}{2}(x + 2) + 3$.

3. Find $\frac{dy}{dx}$ where $y = \frac{\sqrt[3]{x} + 4x^2}{3x^{3/2} + x^3}$

Solution: By the quotient rule,

$$\begin{aligned} y' &= \frac{(\sqrt[3]{x} + 4x^2)'(3x^{3/2} + x^3) - (\sqrt[3]{x} + 4x^2)(3x^{3/2} + x^3)'}{(3x^{3/2} + x^3)^2} \\ &= \frac{(\frac{1}{3}x^{-2/3} + 8x)(3x^{3/2} + x^3) - (\sqrt[3]{x} + 4x^2)(\frac{9}{2}x^{1/2} + 3x^2)}{(3x^{3/2} + x^3)^2} \end{aligned}$$

Find where the function $f(x) = x^2e^x$ is concave up.

Solution: To determine concavity, we need the second derivative.

By the product rule, $f'(x) = (x^2)'e^x + x^2(e^x)' = 2xe^x + x^2e^x$.

We can leave the derivative like this and use the product rule on each term when we take the second derivative, but simplifying can help:

$$f'(x) = e^x(2x + x^2).$$

Then we take the second derivative by the product rule:

$$\begin{aligned} f''(x) &= (e^x)'(2x + x^2) + e^x(2x + x^2)' \\ &= e^x(2x + x^2) + e^x(2 + 2x) = e^x(x^2 + 4x + 2) \end{aligned}$$

We simplify again here so we can solve the problem - since we want to know where the second derivative is greater than 0 to find where f is concave up.

Note that $f''(x) = e^x(x^2 + 4x + 2) > 0$ at the same times when $x^2 + 4x + 2 > 0$ because $e^x > 0$ for all x .

Knowing that the graph of $y = x^2 + 4x + 2$ is a parabola opening up, we can find where this quadratic is equal to zero and know the answer is the sections to the left and the right of the zeros. This quadratic does not factor, so we use the quadratic formula:

$$x = \frac{-4 \pm \sqrt{4^2 - 4(1)(2)}}{2(1)} = \frac{-4 \pm \sqrt{8}}{2} = \frac{-4 \pm 2\sqrt{2}}{2} = -2 \pm \sqrt{2}.$$

Therefore, f is concave up on $(-\infty, -2 - \sqrt{2}) \cup (-2 + \sqrt{2}, \infty)$ (where the $\cup$ symbol means union, the combination of the two intervals.)